American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Fish Disease Detector

A Software Engineering Project Submitted

By

Semester: Summer_23-24		**Section:F**	**Group Number:2**	
SN	Student Name	Student ID	Contribution (CO3+CO4)	Individual Marks
1	**Mahmud Reza Mahim**	**22-46210-1**		
2	**S.M.Sayed Arifin Omi**	**22-46613-1**		
3	**Rangon Kumar Shaha**	**22-46585-1**		
4	**MD.Mahedi hasan Chowdury**	**22-46580-1**		
5	**Aklima Akther Akhi**	**22-46750-1**		

The project will be Evaluated for the following Course Outcomes

CO3: *Select* appropriate software engineering models, project management roles and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects	Total Marks	
Appropriate Process Model Selection and Argumentation with Evidence	[5 Marks]	
Evidence of Argumentation regarding process model selection	[5Marks]	
Analysis the impact of societal, health, safety, legal and cultural issues	[5Marks]	
Submission, Defense, Completeness, Spelling, grammar and Organization of the Project report	[5Marks]	
	Total Marks	

CO4: *Develop* project management plan to manage software engineering projects following the principles of engineering management and economic decision process		
Develop the project plan, its components of the proposed software products	[5Marks]	

Identify all the activities/tasks related to project management and categorize them within the WBS structure. Perform detailed effort estimation correspond with the WBS and schedule the activities with resources	[5Marks]	
Identify all the potential risks in your project and prioritize them to overcome these risk factors.	[5Marks]	

Description of Student's Contribution in the Project work

Student Name: **Mahmud Reza Mahim**
Student ID: **22-46210-1**
Contribution in Percentage (20%):
Contribution in the Project:
Solution to the problem
Functionality
Class Diagram(brain storming)
Warframe design
testcase
testplanning

____

Signature of the Student

Student Name: **S.M. Sayed Arifin Omi**
Student ID: **22-46613-1**
Contribution in Percentage (20%):
Contribution in the Project:
Solution to the problem
Functionality
Activity Diagram
Selection of process model
Warframe design
testcase
testplanning
WBS
Effort estimation & cocomo

Signature of the Student

Student Name: **Rangon Kumar Shaha**
Student ID: **22-46585-1**
Contribution in Percentage (20%):
Contribution in the Project:
Background to the problem
Functionality
UseCase diagram
Class Diagram
Sequence Diagram
Selection of process model
Warframe design
testcase
testplanning
WBS
Effort estimation & cocomo

Signature of the Student

Student Name: **MD.Mahedi hasan Chowdury**
Student ID: **22-46580-1**
Contribution in Percentage (20%):
Contribution in the Project:
Background to the problem & Root analysis
Activity & Sequence Diagram(Brain Storming)
Selection of process model
Warframe design
testcase
testplanning

Signature of the Student

Student Name: **Aklima Akther Akhi**
Student ID: **22-46750-1**
Contribution in Percentage (20%):
Contribution in the Project:
Solution to the problem
Functionality
UseCase diagram(Brainstroming)
Warframe design
testcase
testplanning
WBS

Signature of the Student

Project Proposal:

Background to the problem :

A software solution for fish farms in Bangladesh. It addresses challenges like manual observation and lack of real-time monitoring. With advanced technology, it aims to bolster sustainability and productivity in the fish farming sector.Fish disease detection is crucial in Bangladesh due to its heavy reliance on aquaculture for food security,livelihoods and economic growth With a large population dependent on the sector,disease outbreaks van result in significant socio-economic impacts.While some study has been conducted on fish disease detection in Bangladesh, further study is essential.Limited resources ,rapid environmental changes, and diverse aquaculture practices pose unique challenges that require tailored detection solutions.

The main problem with the current fish disease detection solution in Bangladesh is their reliance on manual observation and visual inspection,which are labor-intensive,subjective,and prone to human error. This approach often results in delayed or inaccurate diagnosis, leading to increased economic losses for farmers. Additionally, the lack of real-time monitoring capabilities hampers early detection and intervention efforts, allowing diseases to spread unchecked and further exacerbating the economic impact on fish farms. Therefore there is a critical need for the development and adoption of software-based fish disease detection solution to address these challenges and improve the economic sustainability of aquaculture in Bangladesh.

Solution to the problem:

The objective of the project is to develop a comprehensive understanding of fish disease detection and management in order to create effective strategies and tools for addressing these challenges. This may involve developing software solutions, conducting research, implementing preventive measures, and providing support to fish farmers to enhance their capacity for disease detection and management. Ultimately the goal is to improve the sustainability, productivity and resilience of the aquaculture sector by minimizing the impact of fish disease on economic, environmental and social factors.

Fish farming, a critical component of aquaculture, is often plagued by challenges related to disease detection and management. Addressing these issues can significantly enhance productivity and sustainability. An innovative solution involves the development of a mobile application designed to assist fish farmers in detecting diseases, recommending appropriate medications, and locating nearby suppliers. This essay outlines the key features, benefits, and implications of such an application, demonstrating its potential to revolutionize fish health management. **Fish Disease Detection**

The cornerstone of the mobile application is its image recognition system, capable of analyzing images of diseased fish captured by users using their smartphone cameras. Leveraging machine learning algorithms, the system will identify common fish diseases based on visual symptoms such as lesions, discoloration, or abnormal behavior. This real-time analysis enables timely interventions, preventing the spread of diseases and reducing economic losses.

Medication Recommendation

Upon detecting a fish disease, the application will recommend appropriate medications or treatments based on established protocols and expert recommendations. The recommendation algorithm will match the detected diseases with the most effective and suitable medications available in the system's database. This feature ensures that fish farmers receive accurate and personalized treatment suggestions, improving the chances of successful disease management.

Nearby Shop Locator

Integrating geolocation services, the application will help users locate nearby shops or pharmacies where they can purchase the recommended medicines. By leveraging mapping APIs and real-time location data, the system provides users with a convenient way to access the necessary medications without extensive searching. This functionality not only saves time but also ensures that treatments can be administered promptly.

Category: The project falls within Category B, indicating the existence of a pre-existing solution. However, our endeavor aims to augment the functionality of current applications within the same domain. Examples of analogous applications include FishVet and DFYOLO.

Our proposed enhancements will include features such as:

User-Friendly Interface: Develop a mobile application that allows fish farmers to easily capture images of diseased fish and receive real-time analysis and recommendations.

Advanced Algorithms: Implement advanced image recognition and machine learning algorithms to accurately identify common fish diseases based on visual symptoms.

Medication Recommendations: Integrate a medication recommendation feature that suggests appropriate treatments for detected diseases, based on established protocols and expert knowledge.

Geolocation Services: Incorporate geolocation services to help users locate nearby shops or pharmacies where they can purchase the recommended medications.

Appropriateness of the Solution

Accessibility

A mobile application offers a convenient and accessible platform for fish farmers to access disease detection and medication recommendation services directly from their smartphones, eliminating the need for specialized equipment or expertise.

Real-Time Analysis

The use of image recognition and machine learning enables real-time analysis of fish health, allowing for timely detection and treatment of diseases to prevent further spread and minimize economic losses.

Customization

By recommending personalized treatment options based on the specific diseases detected in their fish, the application offers tailored solutions that meet the individual needs of each user.

Convenience

The integration of geolocation services simplifies the process of locating nearby shops selling the required medications, eliminating the need for users to search extensively for suppliers.

Feasibility to Meet Business Objectives

The proposed solution aligns with the business objective of providing value-added services to fish farmers, enhancing fish health management practices, and ultimately improving productivity and profitability in aquaculture operations. By offering a comprehensive solution that addresses key challenges faced by fish farmers in disease detection and treatment, the mobile application has the potential to attract a large user base and generate revenue through subscription fees, in-app purchases, or partnerships with pharmaceutical suppliers. The technical feasibility of implementing the proposed solution is high, given advancements in image recognition technology, machine learning algorithms, and geolocation services.

Basic Functionalities of Fish Disease Detection

Early Detection and Monitoring

Advanced sensors and imaging systems continuously monitor fish health parameters in aquaculture facilities. Early detection prevents disease outbreaks, reduces fish mortality, and ensures sustainable production, safeguarding food security and livelihoods.

Machine Learning Algorithms

Machine learning models analyze complex data to predict disease occurrence. Accurate predictions allow timely interventions, minimizing economic losses and promoting safety in aquaculture practices.

Immunohistochemistry and Molecular Techniques

Immunohistochemistry identifies specific pathogens in tissue samples, providing precise diagnosis and informing treatment strategies, ensuring legal compliance with disease control regulations.

Community Engagement and Education

Developing educational programs for fish farmers, consumers, and local communities empowers them with knowledge about disease prevention, safety practices, and legal responsibilities.

Target Group of Users and Benefits

Target Group of Users

- Fish Farmers and Aqua culturists
- Researchers and Scientists
- Aquarium Owners and Hobbyists

Benefits

- Early Disease Detection: Improved fish health through timely detection and intervention.
- Cost Saving: Reduced economic losses due to effective disease management.
- Improved Sustainability: Sustainable aquaculture practices through better disease control.

Contribution to Scientific Knowledge and Documented Results:

Advancement in Fish Health Research

Researchers can use the data collected by the system to study disease prevalence, patterns, and risk factors. Documented results enhance understanding of fish diseases and their impact on aquaculture.

Validation of Detection Algorithms

Documentation of algorithm performance metrics adds to the scientific literature, providing benchmarks for future research.

Case Studies and Real-World Implementation

Case studies provide practical insights and help researchers learn from the project's challenges, successes, and lessons learned.

Data Sharing and Collaboration

Open datasets facilitate collaborative research, contributing to reproducibility and transparency in scientific studies.

Economic and Environmental Impact Assessment

Environmental impact assessments highlight the positive effects of disease control on aquaculture sustainability.

Educational Resources for Future Studies

Sharing best practices for disease detection and management through outreach efforts raises awareness about fish health and promotes responsible aquaculture practices.

The proposed mobile application for fish disease detection and medication recommendation represents a comprehensive solution that addresses the critical needs of fish farmers. By integrating advanced technologies and offering user-friendly features, the application has the potential to significantly improve fish health management, contribute to scientific knowledge, and enhance the overall sustainability and productivity of aquaculture operations.

Functionality:

1. **Software Signup**

- New users can create an account by providing necessary details such as name,email address and password to access the software's features and services.
- The signup credentials will be added in database record.

Priority Level: High

Precondition : User needs valid email and minimum 8 digit password.

2. **Software Login**

- The software shall allow users to login with their given email and password.
- The login credentials (email and password) will be verified with database records

Priority Level: High

Precondition : User must have valid email and password.

3. **Forget Password**

- Users can recover their account password by following the “Forgot Password” process,which typically involves receiving a password reset link or OTP(One-Time Password) via email.
- After clicking on password reset link or entered otp, new window will be showed.There user can set new password.

Priority Level: High

Precondition : User must have email access.

4. **Photo capture:**

- The system shall allow users to upload/take images of fish for analysis.
- The system shall support various image formats including JPEG, PNG, and BMP.

Priority Level: High

Precondition : User have targeted photo.

5. **Disease Detection:**

- The system shall utilise image recognition algorithms to identify potential diseases in Fish.

- The system shall analyse visual cues such as discoloration, lesions, abnormal behaviour and other symptoms to detect diseases.

- The system shall provide a list of possible diseases along with confidence levels for each detection.

Priority Level: High

Precondition : N/A

6. Disease Diagnosis:

- The system shall provide detailed information about detected diseases, including symptoms, causes, and recommended actions.
- The system shall offer references to authoritative sources for further reading and verification.

Priority Level: High

Precondition : N/A

7. User Profile

- User can create and manage their profiles,including detailed information such as fish species cultivated,previous disease history,and specific preferences for disease management strategies.

Priority Level: High

Precondition : The user must be logged in.

8. An assistant powered by AI:

- The software includes an AI-powered assistant specially trained in fish disease detection and management,providing users with guidance, recommendations,and assistance in analysing disease symptoms,interpreting data,and implementing treatment plans.

Priority Level: High

Precondition : Device must be connected to internet

9. Notification Alerts:

- The software sends real-time notification alerts to users regarding disease outbreaks,abnormal trends in fish health parameters,treatment reminders,and important updates related to fish disease detection and management.

Priority Level:Medium

Precondition : N/A

10. Live location tracker:

- The software tracks real time using GPS and shows the nearby medical stores.

Priority Level: High

Precondition: The device supports gps.

11. Live consultation:

- The software will connect users to a real veterinary so that users can get live consultation at any time.

Priority Level: medium

Precondition: At the time of requests a veterinary should be available.

12. Help line:

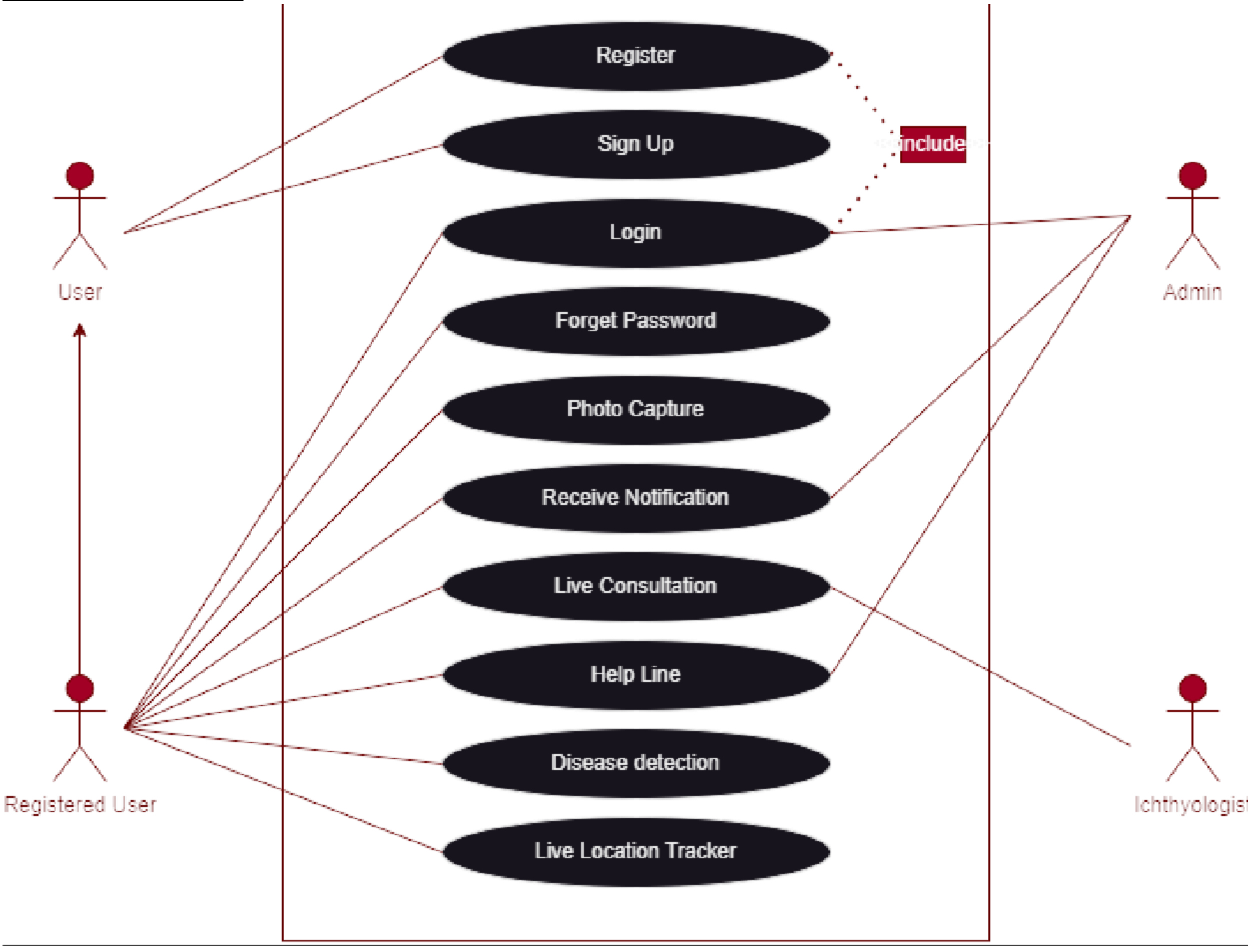
- Allows users to get any type of real time software support.

Priority Level: Low

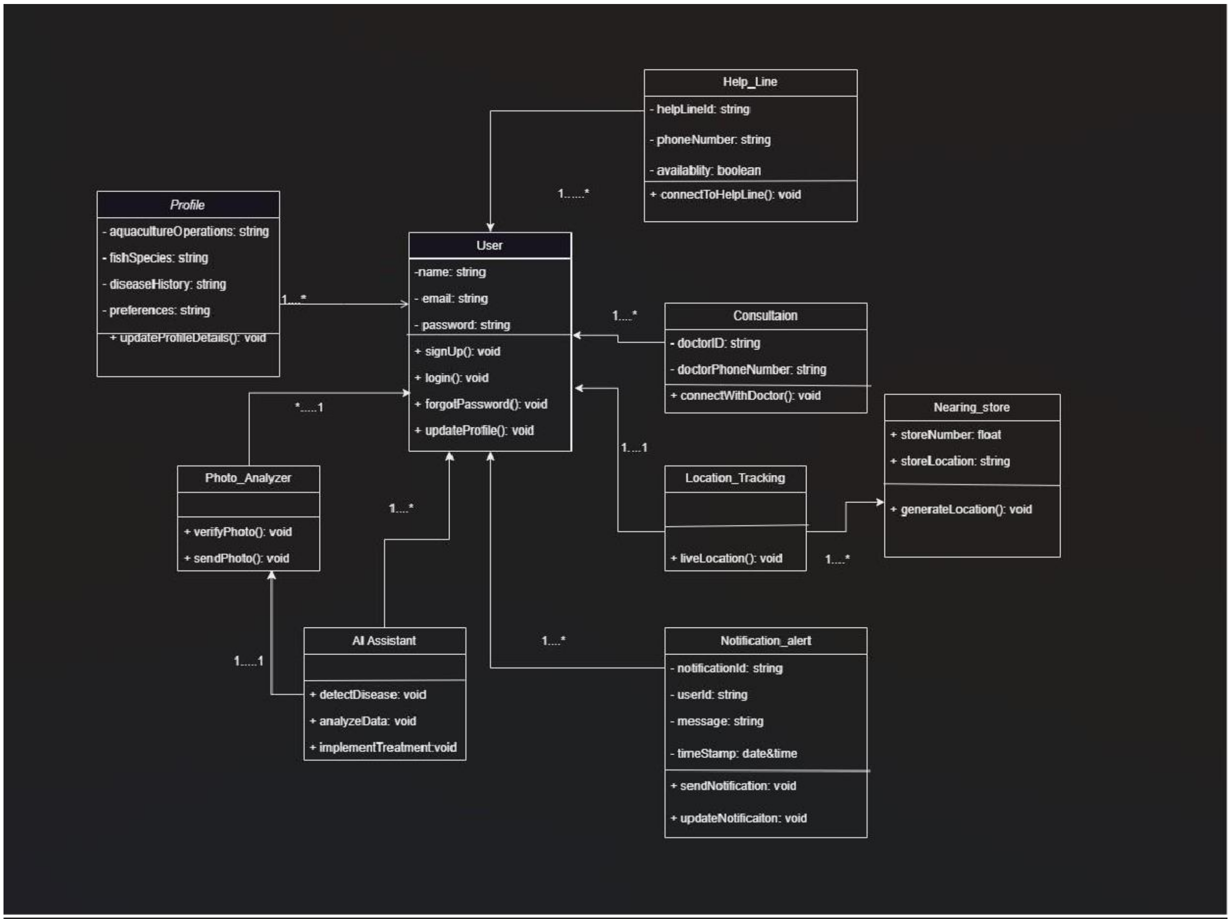
Precondition: At the time of requests a customer service agent should be available.

Diagram:

Usecase diagram:

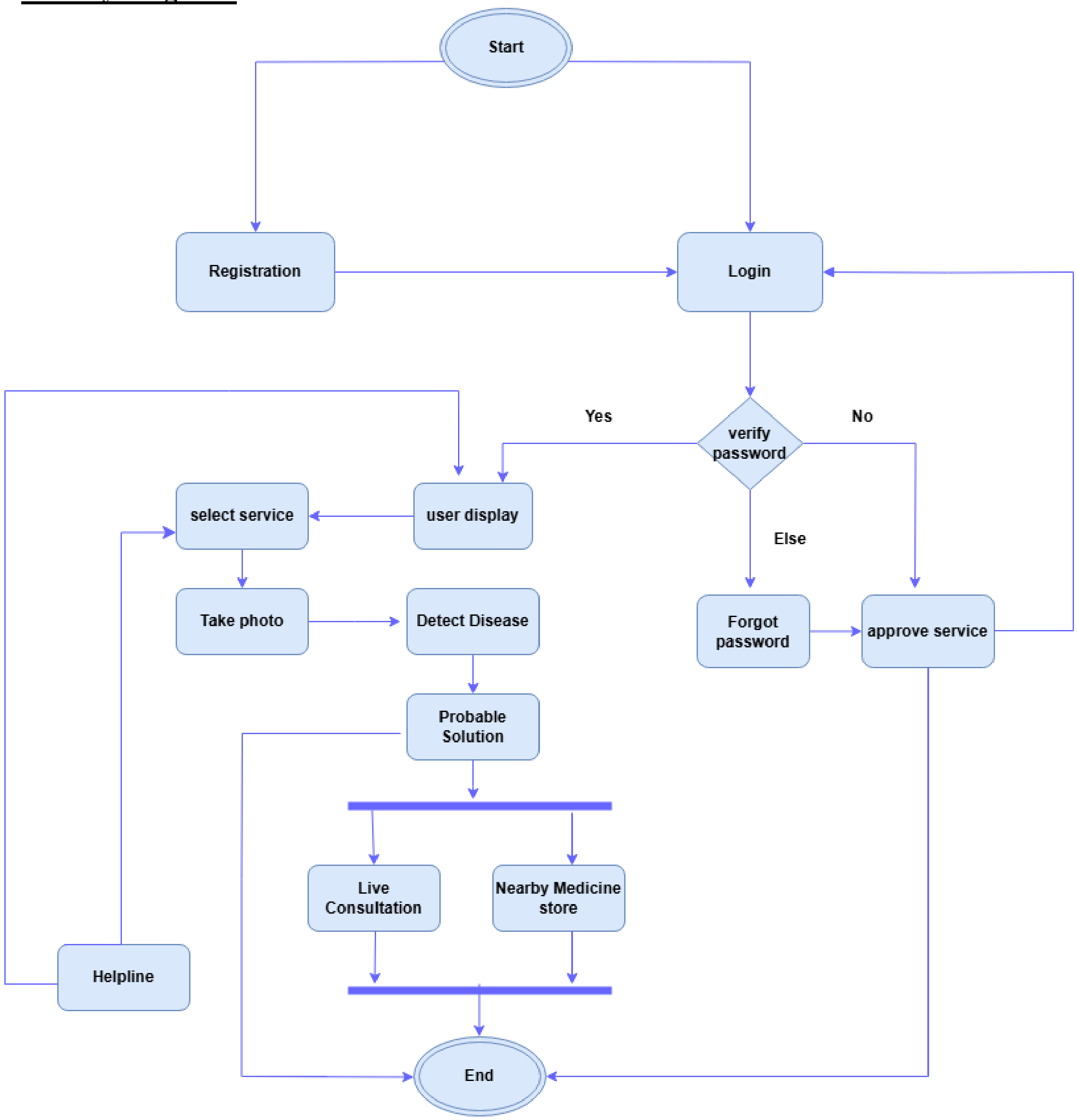


Class Diagram:

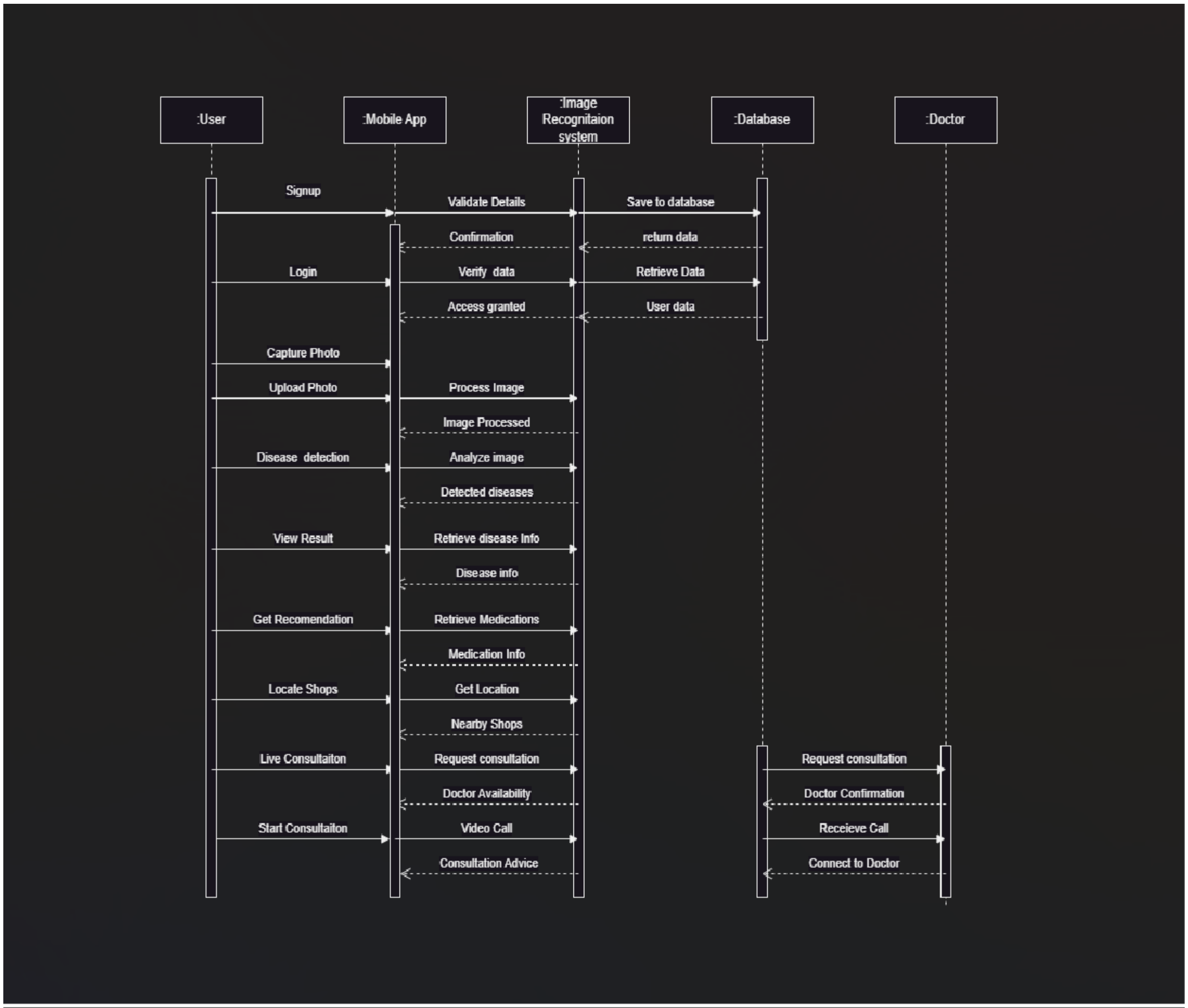


© MMH

Activity Diagram:



Sequence Diagram:



Selection of Process Model:

We use the Incremental Process Model to develop this project because it provides flexibility, accommodates evolving requirements, manages risks effectively, ensures early delivery of value, and promotes continuous improvement based on stakeholder feedback. These benefits are essential for developing a complex software application where accuracy, reliability, and user satisfaction are critical factors for success. Therefore, the Incremental Process Model aligns well with the dynamic nature of our project and supports its overall goals of delivering a robust and effective solution for fish disease detection and management. In software development project management, several roles play crucial parts in ensuring the successful planning, execution, and delivery of software products. These roles have distinct responsibilities that contribute to different aspects of the project lifecycle. Here are the key roles in project management activities in software development along with their responsibilities:

1. **Project Manager**

   Responsibilities:

   - Overall responsibility for the planning, execution, monitoring, controlling, and closure of a project.
   - Develop and manage project schedules and timelines.
   - Communicate project status, issues, and risks to stakeholders.
   - Manage project budget and ensure financial accountability.
   - Facilitate team meetings, decision-making processes, and conflict resolution.
   - Ensure adherence to project management methodologies and best practices.

2. **Product Owner**

   Responsibilities:

   - Represents stakeholders and customers to the development team.
   - Defines and prioritizes product features and requirements.
   - Manages and maintains the product backlog.
   - Works closely with stakeholders to gather requirements and feedback.
   - Accepts or rejects work results based on acceptance criteria.
   - Provides direction and vision to the development team regarding the product.

3. **Scrum Master (if using Agile/Scrum methodology)**

   Responsibilities:

   - Facilitates Agile ceremonies (sprint planning, daily stand-ups, sprint reviews, retrospectives).
   - Removes impediments that hinder the team's progress.

- Ensures the team adheres to Agile principles and practices.
- Protects the team from external distractions.
- Helps the team improve continuously.

4. **Development Team (Developers, Testers, Designers)**

Responsibilities:

- **Developers:** Write, test, and maintain code that meets product requirements.
- **Testers:** Create and execute test cases to ensure software quality and functionality.
- **Designers:** Create user interfaces, graphics, and overall user experience design.
- Collaborate effectively within the team and with other stakeholders.

5. **Quality Assurance (QA) Manager/Engineer**

Responsibilities:

- Develops QA strategies and test plans for the project.
- Coordinates and performs testing activities to identify defects.
- Ensures software quality through systematic testing and validation.
- Manages the QA team and resources.
- Provides metrics and reports on testing progress and quality assurance status to stakeholders.
- Collaborates with the development team to implement testing strategies and continuous improvement processes.

6. **Technical Lead/Architect**

Responsibilities:

- Defines the overall architecture and technical design of the software system.
- Guides and mentors the development team on technical aspects and best practices.
- Ensures that the architecture supports scalability, performance, and maintainability goals.
- Conducts code reviews and ensures adherence to technical standards and guidelines.
- Collaborates with stakeholders to understand requirements and translate them into technical solutions.
- Leads technical discussions, problem-solving sessions, and decision-making processes.

7. **Stakeholders and Users**

Responsibilities:

- Provide input on requirements, priorities, and acceptance criteria.
- Participate in reviews, demos, and acceptance testing.
- Provide feedback on the product throughout its development lifecycle.
- Ensure that the final product meets business needs and user expectations.
- May include end-users, customers, sponsors, and other relevant parties.

8. **System Administrator/Operations Team**

Responsibilities:

- Prepare for the deployment and release of the software.
- Manage the infrastructure and environment where the software will run.
- Provide ongoing support and maintenance after deployment.
- Monitor system performance and respond to incidents or issues.
- Ensure smooth operations and uptime of the software product.

Flexibility and Adaptability:

- The Incremental Process Model offers flexibility to accommodate changing requirements, priorities, and stakeholder feedback throughout the project lifecycle.
- This flexibility allows for continuous improvement and adjustment, ensuring that the final product meets evolving needs effectively.

Early and Continuous Delivery of Value:

- The Incremental Process Model emphasizes delivering working software increments early and frequently.
- Each increment focuses on delivering a subset of features or functionality, allowing stakeholders to see tangible progress and provide feedback early in the process.
- Early delivery of functional increments enables validation of requirements and ensures alignment with user expectations.

Risk Management and Mitigation: • Incremental development includes iterative planning, development, testing, and review processes.

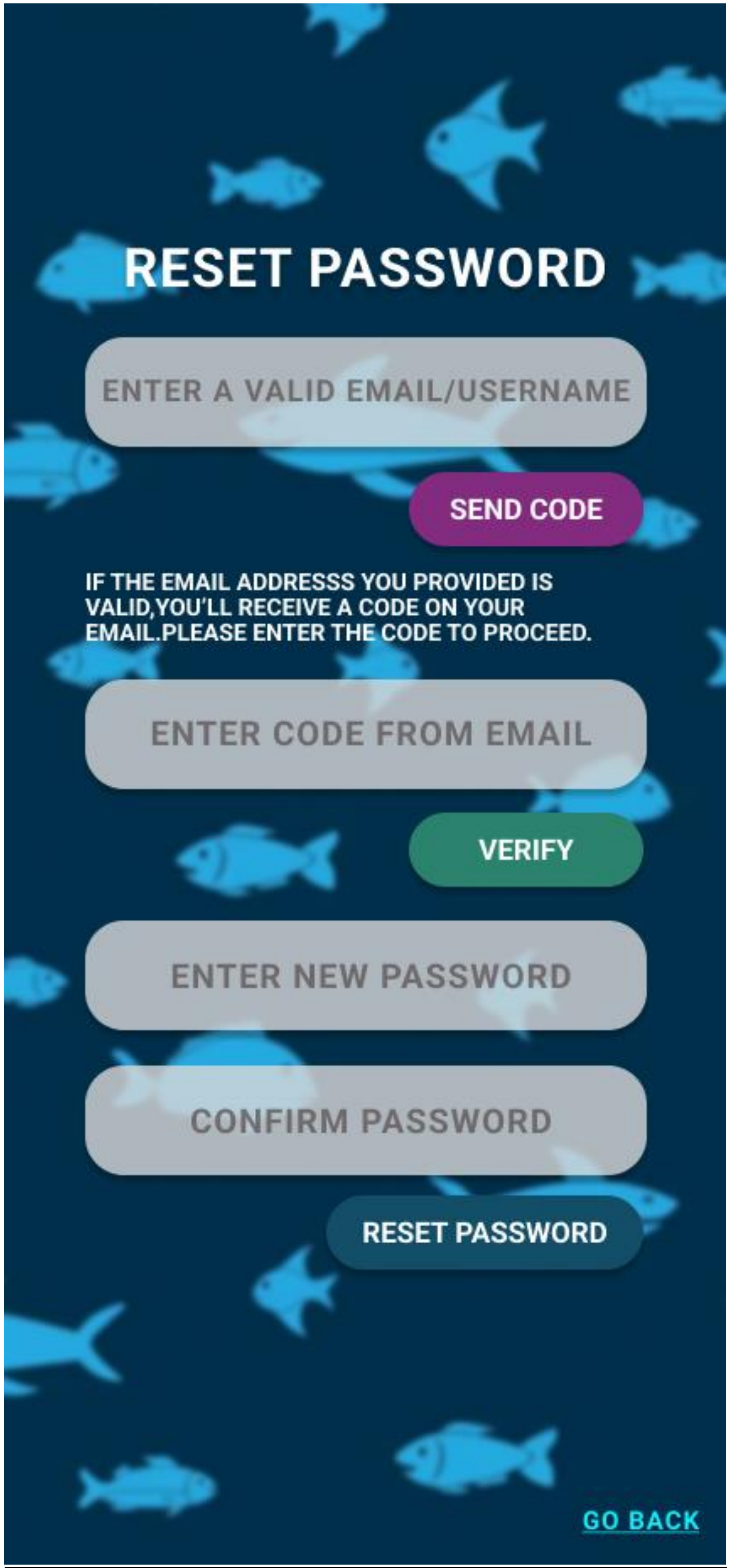
- Risks are identified and addressed incrementally, reducing the impact of potential issues and enhancing project predictability and control.
- Issues and challenges can be addressed early in the process, minimizing their impact on overall project timelines and outcomes.

Enhanced Stakeholder Engagement: • Incremental models promote continuous stakeholder engagement throughout the development cycle.

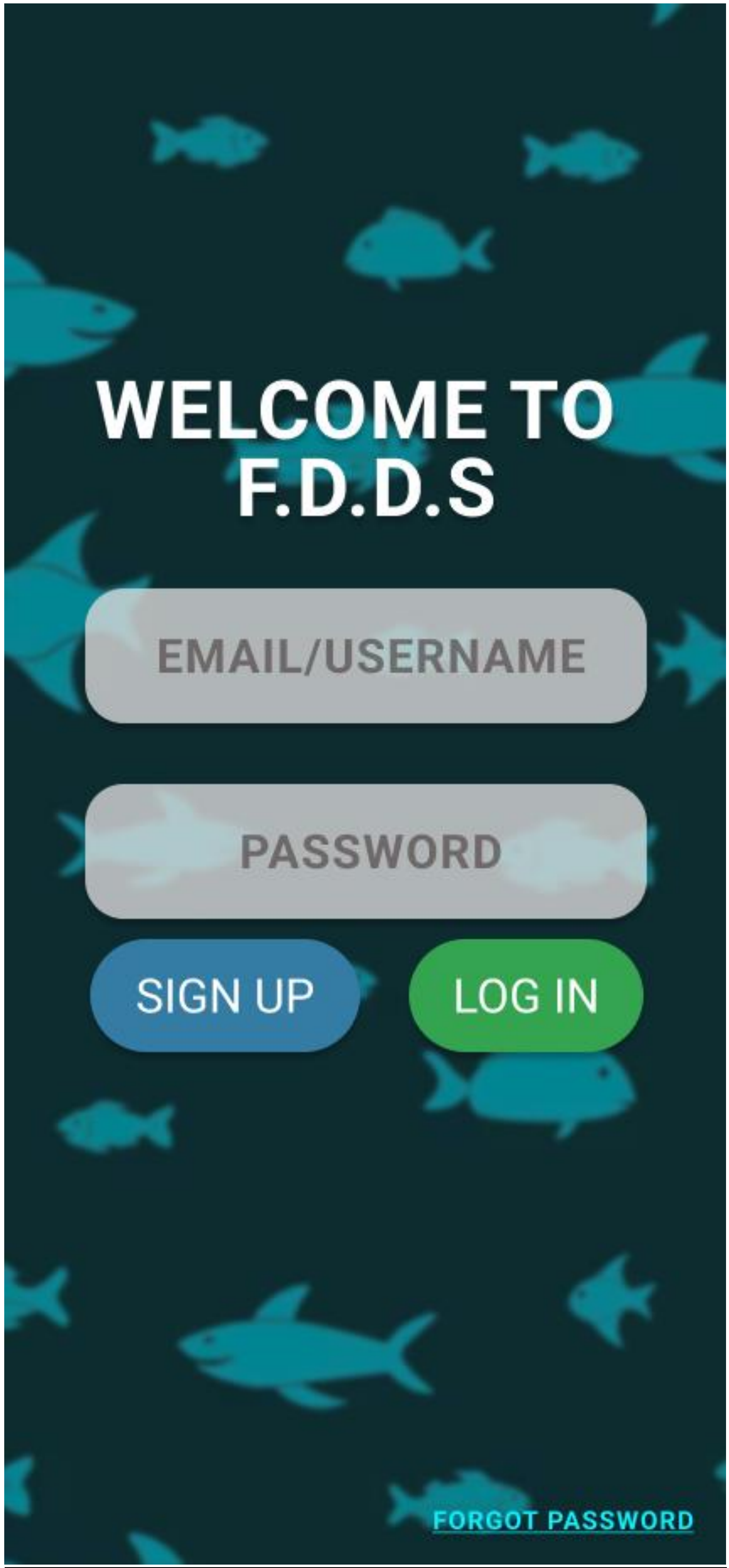
- Regular feedback loops allow stakeholders to provide input on evolving requirements, priorities, and features.
- Stakeholder involvement fosters transparency, collaboration, and ensures that the final product meets business objectives effectively.

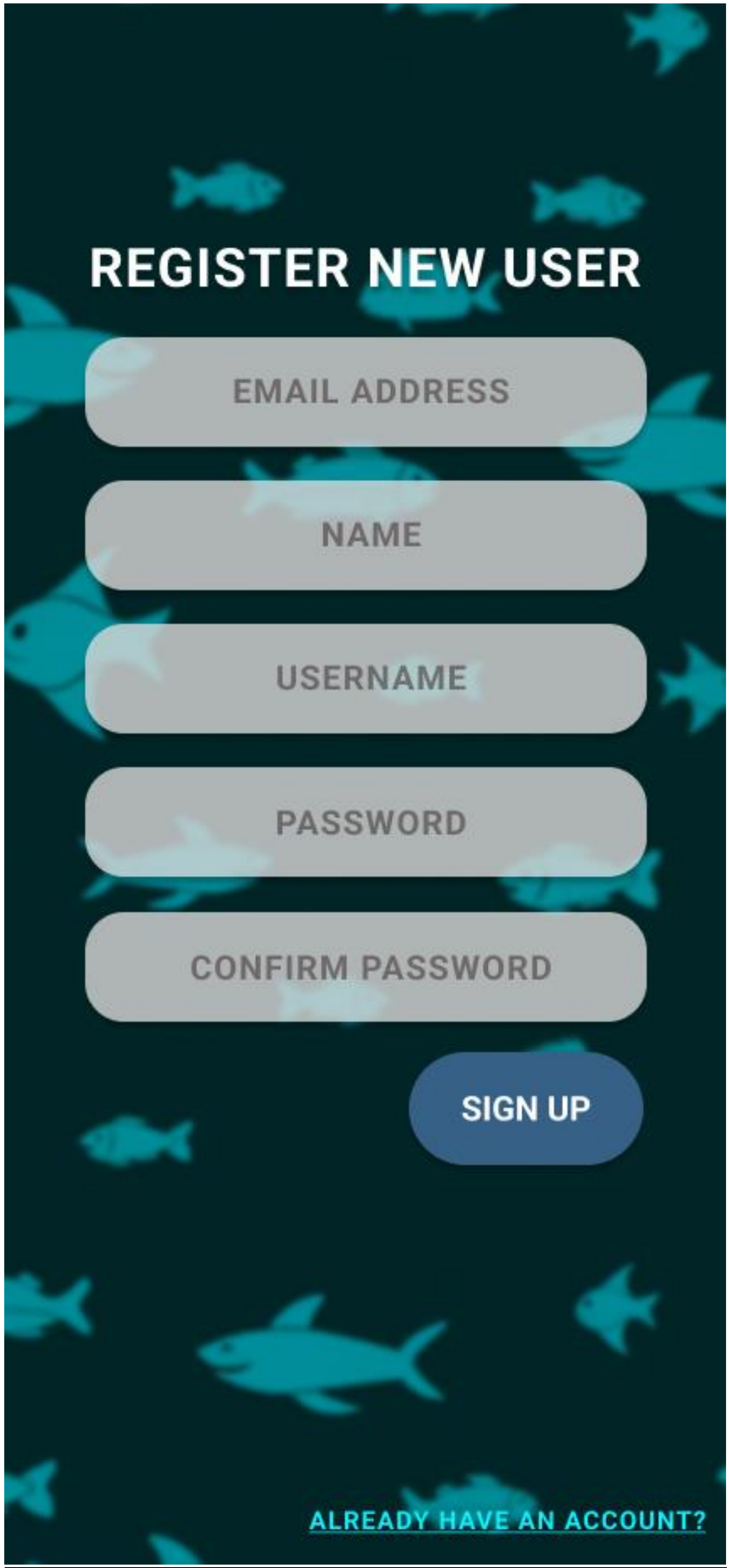
Continuous Improvement and Adaptation: • Incremental development encourages a culture of continuous improvement and adaptation.

- Each increment incorporates lessons learned from previous increments, allowing for refinements in design, functionality, and user experience.
- The incremental refinement process helps in achieving a high-quality product that aligns closely with stakeholder expectations and market needs.

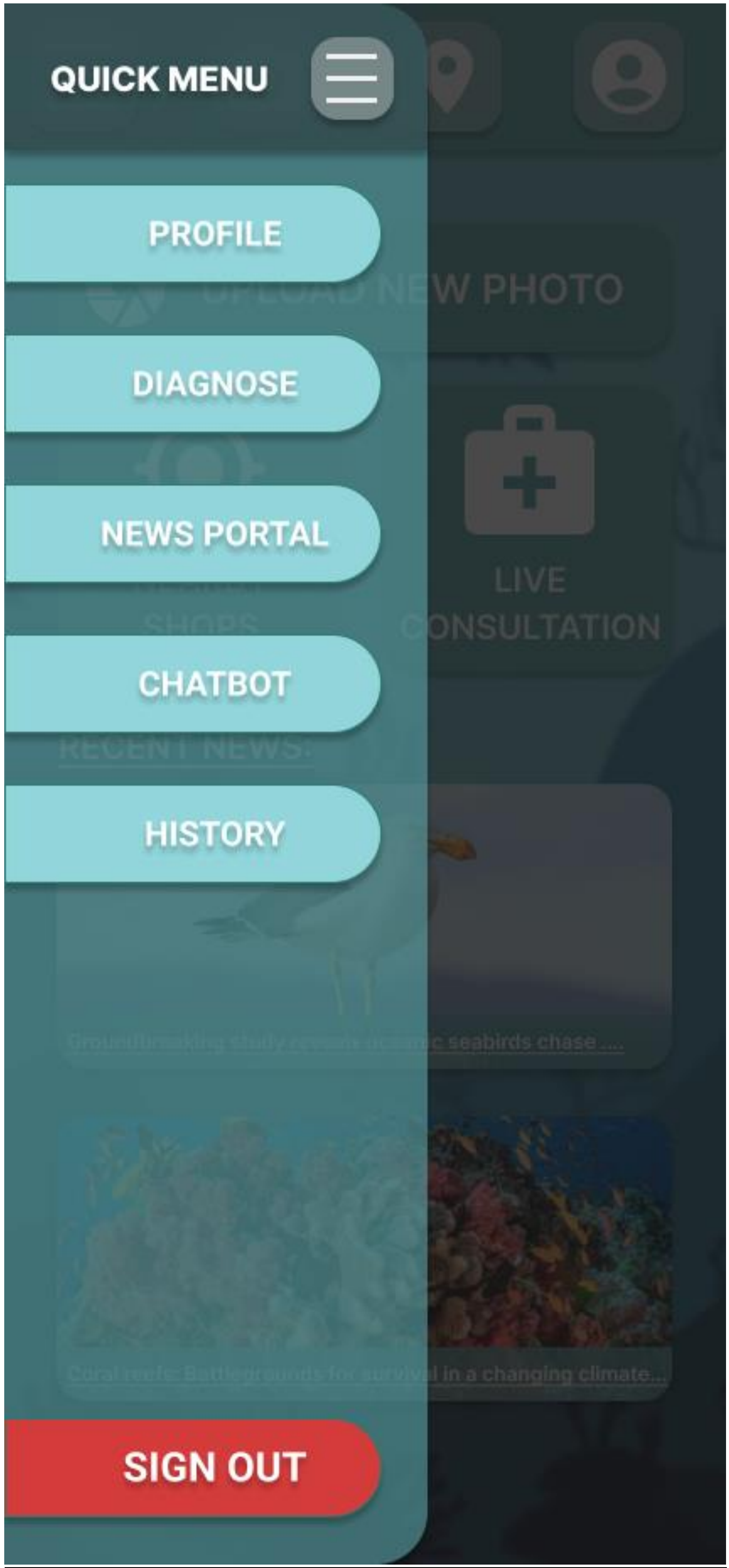

RESET PASSWORD
ENTER A VALID EMAIL/USERNAME
SEND CODE
IF THE EMAIL ADDRESSS YOU PROVIDED IS VALID,YOU'LL RECEIVE A CODE ON YOUR EMAIL.PLEASE ENTER THE CODE TO PROCEED.
ENTER CODE FROM EMAIL
VERIFY
ENTER NEW PASSWORD
CONFIRM PASSWORD
RESET PASSWORD
GO BACK

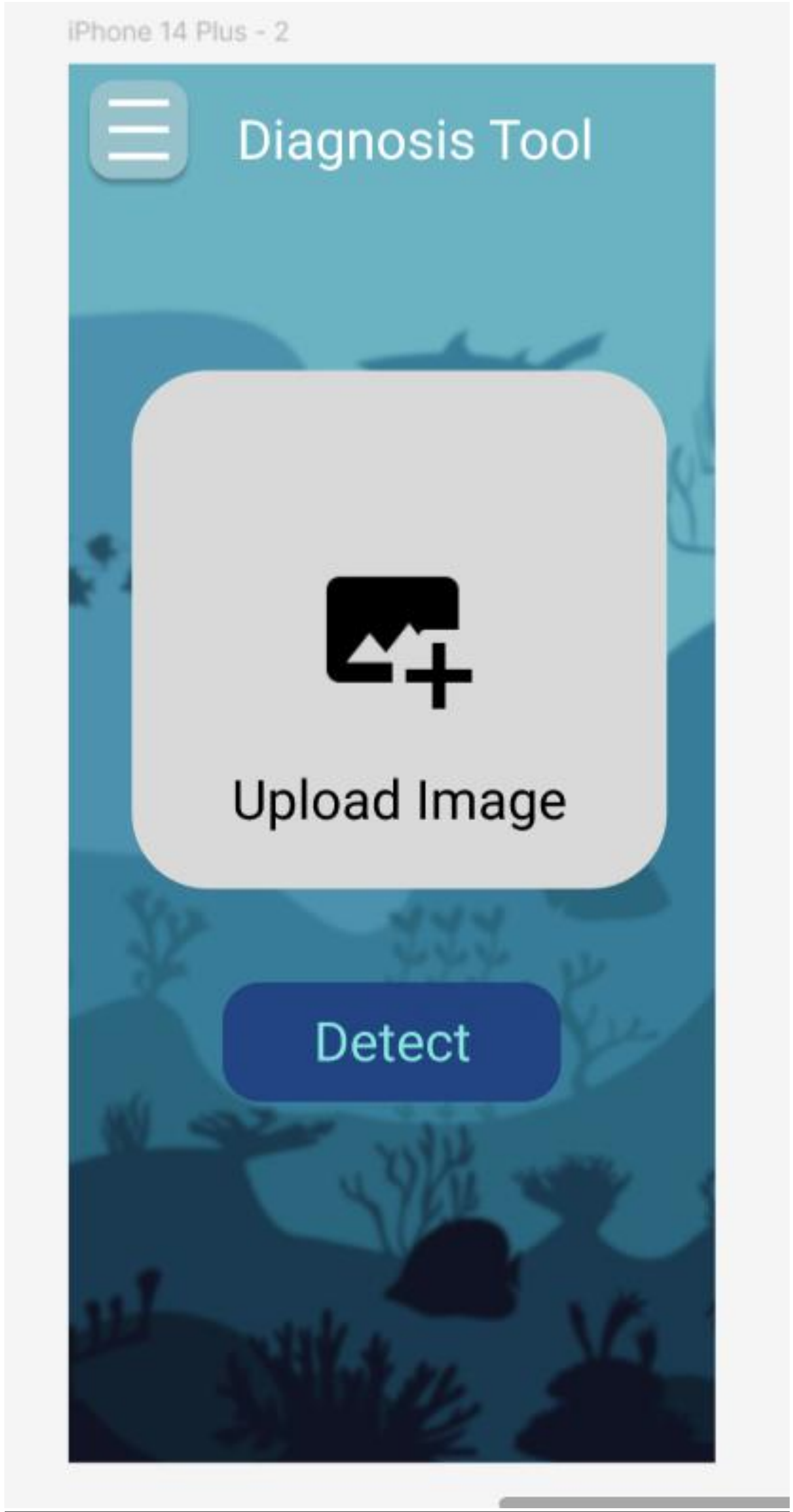

UPLOAD NEW PHOTO
NEARBY SHOPS
LIVE CONSULTATION
RECENT NEWS:
Groundbreaking study reveals oceanic seabirds chase ....
Coral reefs: Battlegrounds for survival in a changing climate...


WELCOME TO
F.D.D.S
EMAIL/USERNAME
PASSWORD
SIGN UP
LOG IN
FORGOT PASSWORD


REGISTER NEW USER
EMAIL ADDRESS
NAME
USERNAME
PASSWORD
CONFIRM PASSWORD
SIGN UP
ALREADY HAVE AN ACCOUNT?

© MMH


QUICK MENU
PROFILE
DIAGNOSE
NEWS PORTAL
CHATBOT
HISTORY
SIGN OUT


iPhone 14 Plus - 2
Diagnosis Tool
Upload Image
Detect


Profile
Sayed Arifin
Name: Sayed Arifin
Email: sayed07@gmail.com
Phone:+88017*****06
Occupation: Doctor
Save Changes


Profile
Sayed Arifin
Name: Sayed Arifin
Email: sayed07@gmail.com
Phone:+88017*****06
Occupation: Doctor
Edit Profile


AI Chatbox
Hello, I am your AI assistant
How can I help you sir?
I want to know about fish disease
Type Here...


Live Location
N
W
E
S
COMPASS
TAGLINE HERE
Location
Compass
Star
Share

Test Case:

Project Name: Fish Disease Detection System	Test Designed by: Rangon Kumar Shaha
Test Case ID: 01	Test Designed date: 2/09/2024
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Diagnosis and analysis	Test Execution date:
Test Title: Diagnosis and Analysis of Fish Diseases	
Description: This test case validates the functionality of diagnosing and analyzing fish diseases using the Diagnosis Tool Page within the Fish diseases detector application. The Diagnosis Tool Page enables users to upload images of affected fish specimens for comprehensive disease analysis.	
Preconditions: 1.User must be logged in to the Fish diseases detector application. 2.Access to the Diagnosis Tool Page available. 3.Fish specimen image(s) uploaded for analysis.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1.Navigate to the Diagnosis Tool Page within the Fish diseases detector application. 2.Upload a fish specimen image for analysis. 3.Initiate the analysis process by selecting the uploaded image. 4.Verify that the analysis is performed promptly and accurately. 5.Review the diagnosis results displayed on the Diagnosis Result Page. 6.Ensure that the diagnosis results provide detailed information about detected fish diseases. 7.Evaluate the accuracy of the diagnosis results based on known fish health conditions. 8.Navigate to the Diagnosis Result Next Page, if applicable, to explore additional disease analysis. 9.Confirm that the Diagnosis Result Next Page extends the analysis with comprehensive insights into fish health.	Fish specimen image(s) for analysis.	1.Diagnosis Tool Page accessible. 2.Successful image upload. 3.Prompt and accurate analysis. 4.Detailed diagnosis results displayed. 5.Diagnosis accuracy confirmed. 5.Additional analysis available if needed.		
Post Condition: Users receive comprehensive analysis and diagnosis of fish diseases based on uploaded images within the Fish diseases detector application's Diagnosis Tool Page.				

Project Name: Fish Disease Detection System	Test Designed by: Rangon Kumar Shaha
Test Case ID: 02	Test Designed date: 2/09/2024
Test Priority (Low, Medium, High): Low	Test Executed by:
Module Name: Menu Page	Execution date:
Test Title: Navigation and Functionality of Menu Page	
Description: This test case verifies the navigation and functionality of the Menu Page within the Fish diseases detection application, which serves as a centralized hub for accessing various features and functionalities.	
Preconditions: 1.User logged into Fish diseases detection.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Open the Fish diseases detection application. Navigate to the Menu Page. 2.Review the available options and features listed on the Menu Page. 3.Click on each option to ensure they lead to the corresponding functionalities. 4.Verify that all options are clickable and functional. 5.Test the responsiveness of the Menu Page across different devices and screen sizes. 6.Ensure that the Menu Page layout is intuitive and user-friendly.	N/A	1.The Fish diseases detection application is accessible. 2.The Menu Page is displayed without errors or delays. 3.All options and features listed on the Menu Page are visible and legible. 4.Clicking on each option leads to the corresponding functionality or page. 5.The Menu Page is responsive across different devices and screen sizes. 6.The layout of the Menu Page is intuitive and promotes seamless navigation.		

Post Condition: Users can easily navigate and access various features and functionalities of the Fish diseases detector application through the Menu Page, enhancing user engagement and experience.

Project Name: Fish Disease Detection System			Test Designed by: S.M.Sayed Arifin Omi	
Test Case ID: 03			Test Designed date: 2/09/2024	
Module Name: Forgot Password			Test Executed by:	
Test Title: Password recovery			Execution date:	
Description: Test forgot the page				
Precondition (If any): User must have a registration account				
Test Steps	Test Data	Expected Results	Actual Resuls	Status (Pass/Fail)
1. Go to Software 2. Go to the Forgot page 3. Enter number 4. Send OTP 5. Enter new password 6. Enter retype password 7. Click login button	User Location	After completing the steps for resetting the password,the user should successfully regain access to their account with the newly set password.		
Post Condition: The user’s password is successfully updated in the system,allowing them to log in to the application using the new password.Additionally,the user should receive a confirmation message or notification indicating that the password reset process was successful.				

Project Name: Fish Disease Detection System	Test Designed by: S.M.Sayed Arifin Omi
Test Case ID: 4	Test Designed date: 09/02/2024
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Registration Session	Test Execution date:
Test Title: Verify provided information and save to database	
Description: Test application Registration page	
Precondition (If any): User must have valid email	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Application 2. Click on Register 3. Enter Email, Password,Mobile and Username.	Email:sayed07@gmail.com Password: 123456 Mobile: 01745424534 Username: Sayed arifin	User should be registered		

Post Condition: The user's registration information is validated and stored in the database, allowing the user to log into the Application with their newly created account.

<table>
<tr><td colspan="3">Project Name:Fish Disease Detection System</td><td colspan="2">Test Designed By:Mahmud Reza Mahim</td></tr>
<tr><td colspan="3">Test Case ID:</td><td colspan="2">Test Designed date: 02/09/24</td></tr>
<tr><td colspan="3">Test Priority (Low, Medium, High): High</td><td colspan="2">Test Executed by:</td></tr>
<tr><td colspan="3">Module Name: Profile Management</td><td colspan="2">Test Execution date:</td></tr>
<tr><td colspan="5">Test Title: Edit Profile Information

Description: Verify that functionalities in Profile management works and profile information can be updated as per user.</td></tr>
<tr><td colspan="5">Precondition (If any): User must have a valid and active profile.</td></tr>
<tr><td>Test Steps</td><td>Test Data</td><td>Expected Results</td><td>Actual Results</td><td>Status (Pass/Fail)</td></tr>
<tr><td>1.Navigate to user profile page

2.Click on the “Edit Profile” Button

3.Update any on the following user information:
• Profile Image
• Profile Name
• Email Address
• Phone Number
• Occupation
• Address
4. Save The updated info.

5.Verify that the profile information is updated .</td><td>Existing Information:
Profile picture:N/A
Profile Name: Sayed Arifin
Email Address:sayed07@gmail.com
Phone:01712565677
Occupation:Doctor

Updated Info:
Profile picture: “profile.png”
Profile Name: Mahmud Reza
Email Address:mahmudrezamahim@gmail.com
Phone:01925258943
Occupation:Farmer</td><td>1.The profile page should display the existing user’s information correctly.

2.After clicking on edit all the fields should be editable.

3.All the information should be updated correctly.

4.The profile page should show the updated information.</td><td></td><td></td></tr>
</table>

Post Condition: User data should be updated in the database.

Project Name:Fish Disease Detection System			Test Designed By:Mahmud Reza Mahim	
Test Case ID:			Test Designed date: 02/09/24	
Test Priority (Low, Medium, High): High			Test Executed by:	
Module Name: AI Powered Assistant			Test Execution date:	
Test Title: AI Chatbot Assistant				
Description: Verify that the AI assistant chatbot works Properly.				
Precondition (If any): User must have be logged in to the software. User must have access to the internet for the AI Assistant to work properly.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Navigate to AI chatbot page 2.Initiate a conversation with the AI Assistant. 3.Ask Various queries related to fish disease. 4.Check chatbot's ability to handle complicated and multiple queries. 5.Ensure Ai bots response follows continuity details.	Various Queries on existing fish diseases. Queries from any existing case study.	The AI assistant should offer assistance related to the queries. Offer veterinarian appointment in emergency/complex case. Offer further assistance		

7.Ensure AI bot offers veterinarian appointments in complex cases. 6.Confirm AI bots offers further assistance.				
Post Condition: The AI Assistance should provide tailored support and guidance.				

<table>
<tr><td colspan="4">Project Name:Fish Disease Detection System</td><td>Test Designed By: Aklima Akther Akhi</td></tr>
<tr><td colspan="4">Test Case ID:</td><td>Test Designed date: 02.09.2024</td></tr>
<tr><td colspan="4">Test Priority (Low, Medium, High): Low</td><td>Test Executed by:</td></tr>
<tr><td colspan="4">Module Name: News Portal</td><td>Test Execution date:</td></tr>
<tr><td colspan="5">Test Title: Verify news content display

Description: Test the functionality of the News Portal to ensure timely and relevant news content is displayed.</td></tr>
<tr><td colspan="5">Precondition (If any): User must be logged in to the application.
User must have an active internet connection.</td></tr>
<tr><td>Test Steps</td><td>Test Data</td><td>Expected Results</td><td>Actual Results</td><td>Status (Pass/Fail)</td></tr>
<tr><td>1.Navigate to the Homepage of the application.

2.Check if the page loads successfully.

3.Verify that news articles displayed in the Recent News section are related to fish disease.</td><td>N/A</td><td>1.The recent news section in homepage loads without errors.

2.Relevant news articles about fish diseases are displayed.</td><td></td><td></td></tr>
<tr><td colspan="5">Post Condition: The Recent News section loads without errors.

Relevant news articles about fish diseases are displayed.</td></tr>
</table>

<table>
<tr><td colspan="3">Project Name:Fish Disease Detection System</td><td colspan="2">Test Designed By:
Aklima Akther Akhi </td></tr>
<tr><td colspan="3">Test Case ID:</td><td colspan="2">Test Designed date:
02.09.2024</td></tr>
<tr><td colspan="3">Test Priority (Low, Medium, High): High</td><td colspan="2">Test Executed by:</td></tr>
<tr><td colspan="3">Module Name: Upload Picture</td><td colspan="2">Test Execution date:</td></tr>
<tr><td colspan="5">Test Title: Image Upload in Diagnosis Tool Page

Description: This test case verifies the functionality of uploading images of affected fish specimens in the Diagnosis Tool Page of the application. The Diagnosis Tool Page allows users to upload images for analysis and diagnosis of fish diseases.</td></tr>
<tr><td colspan="5">Precondition (If any): 1. User must be logged in to the application.
2. Access to the Diagnosis Tool Page available.</td></tr>
<tr><td>Test Steps</td><td>Test Data</td><td>Expected Results</td><td>Actual Results</td><td>Status (Pass/Fail)</td></tr>
<tr><td>1.Navigate to the Diagnosis Tool Page within the application.

2.Locate the image upload feature.

3.Click or tap on the image upload button.

4.Choose a fish specimen image from the local device or camera.

5.Initiate the upload process by selecting the image.

6.Verify the uploaded image is shown in the diagnosis tool.

7.Confirm the picture is uploaded and analysed by the software.</td><td>Any Live image of a sick/healthy fish.

Any image non related to fish disease.</td><td>1.The diagnosis tool can be accessible within the application.

2.The image upload feature is displayed and functional.

3.User can select and upload any image directly from heir camera or gallery.

4.If the image is not related to fish an error occurs and user is redirected to upload image again.

5.The uploaded image is displayed and analyzed.</td><td>As expected.</td><td>Pass</td></tr>
</table>

8.Upload an image not related to fish to ensure it doesn’t false detect information.				

Testing Plan:

This testing plan ensures that all functionalities of the Fish Disease Detection System application are thoroughly validated to meet the specified requirements and quality standards.

- **Start Date:** Monday, Septepber 9, 2024
- **End Date:** Friday, October 25, 2024

Testing Schedule:

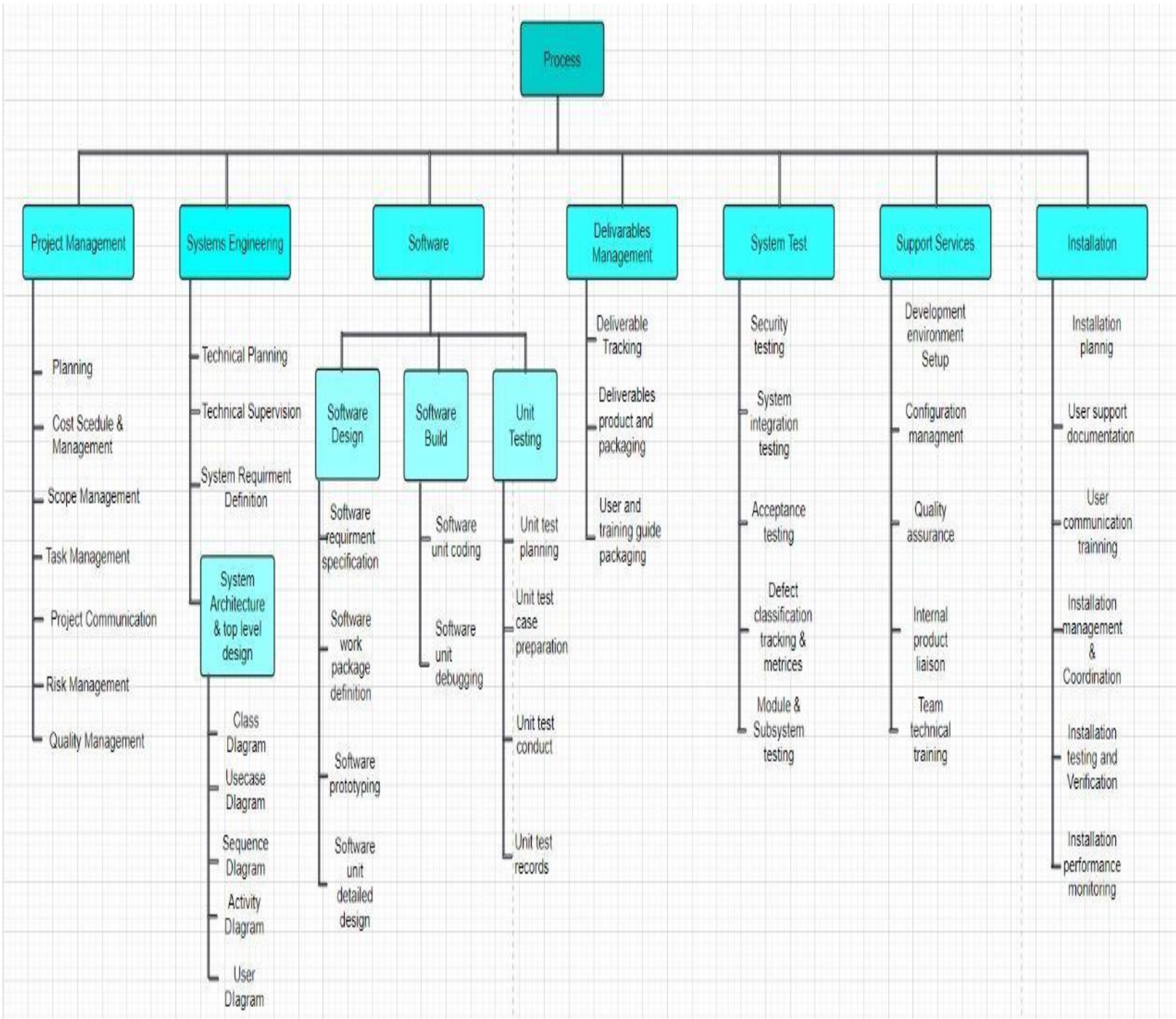
1. **System Testing**
   - **Timeline:** September 9, 2024 - September 13, 2024
   - **Approach:** Conduct end-to-end testing of the system to verify that all integrated components work together as expected.
   - **Responsible Role:** QA Lead
2. **Module Testing: Diagnosis and Analysis**
   - **Timeline:** September 16, 2024 - September 18, 2024
   - **Approach:** Test the Diagnosis Tool Page for accurate detection and analysis of fish diseases using uploaded images.
   - **Responsible Role:** Test Engineer
3. **Module Testing: Menu Page**
   - **Timeline:** September 19, 2024 - September 20, 2024
   - **Approach:** Verify navigation and responsiveness of the Menu Page across different devices.
   - **Responsible Role:** UI/UX Tester
4. **Module Testing: Forgot Password**
   - **Timeline:** September 23, 2024 - September 24, 2024
   - **Approach:** Validate the password recovery process including OTP verification and password reset functionality.
   - **Responsible Role:** Security Tester
5. **Module Testing: Registration Session**
   - **Timeline:** September 25, 2024 - September 26, 2024
   - **Approach:** Test the registration process to ensure accurate data entry and storage in the database.
   - **Responsible Role:** Functional Tester
6. **Integration Testing**
   - **Timeline:** October 1, 2024 - October 4, 2024
   - **Approach:** Test the interaction between integrated modules to confirm data flow and interoperability.
   - **Responsible Role:** QA Lead
7. **User Acceptance Testing (UAT)**
   - **Timeline:** October 7, 2024 - October 11, 2024
   - **Approach:** Conduct UAT to gather feedback from end-users and validate the system against real-world use cases.
   - **Responsible Role:** UAT Coordinator

Tester Roles and Module Assignments:

- **QA Lead**
    - **Responsibilities:** Oversee the testing process, ensure all test cases are executed, and validate overall system performance.
    - **Assigned Modules:** System Testing, Integration Testing
- **Test Engineer**
    - **Responsibilities:** Execute detailed module-level tests to ensure specific functionalities meet requirements.
    - **Assigned Modules:** Diagnosis and Analysis, AI Powered Assistant
- **UI/UX Tester**
    - **Responsibilities:** Validate user interface and experience across various devices and screen sizes.
    - **Assigned Modules:** Menu Page, Profile Management
- **Security Tester**
    - **Responsibilities:** Focus on testing security features such as authentication and data protection.
    - **Assigned Modules:** Forgot Password, Registration Session
- **Functional Tester**
    - **Responsibilities:** Conduct functionality checks to verify all features perform as expected.
    - **Assigned Modules:** News Portal, Image Upload in Diagnosis Tool Page
- **UAT Coordinator**
    - **Responsibilities:** Facilitate User Acceptance Testing by coordinating with end-users and collecting feedback.
    - **Assigned Modules:** All Modules for UAT

This plan ensures a comprehensive evaluation of all critical functionalities within the Fish Disease Detection System application.

<u>Work Breakdown Structure(WBS):</u>



Effort Estimation:

LOC(Line of Code) = 6000
Special Skill (B) = 2
Productivity Parameter (P) = 1.5
Project Duration (t) = 4 months

Now,
$E=(6000*2^{0.333}/1.5)^3 *(¼)^4$
$=499.65*10^6$

COCOMO:

As our project is an organic software,
Therefore:
Coefficient = 2.4
Project complexity (P) = 1.05
SLOC-dependent coefficient (T) = 0.38

1. Effort calculation:

$PM=Coefficient \times(SLOC /1000)^P = 2.4 \times (6000/ 1000)^{1.05}$
$=15.75$ person-months

2.Development Time:

$DM=2.50\times (PM)^T = 2.50 \times (15.75)^{0.38} = 7.13$ months

3. Required number of People:

$ST = PM/DM= 15.75/ 7.13 =2.21 \approx 3$ people